

Lecture 5: Calculus and Linear Algebra

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Course content (math tapas)

Sequences
and series

Functions
of one variable

Differentiation

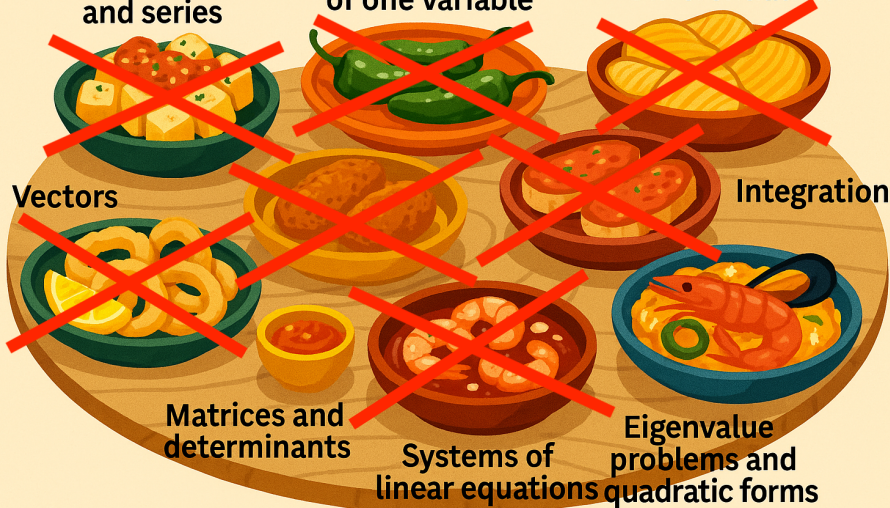
Vectors

Integration

Matrices and
determinants

Systems of
linear equations

Eigenvalue
problems and
quadratic forms



Spectral decomposition and applications

Spectral decomposition (symmetric case)

Theorem (Spectral Theorem)

If $A \in \mathbb{R}^{n \times n}$ is symmetric, then there exist an orthogonal $U \in \mathbb{R}^{n \times n}$ and a diagonal $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that

$$A = U\Lambda U^\top, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n.$$

The columns of U are orthonormal eigenvectors of A .

Application: Principal Component Analysis (PCA)

We observe n data points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$.

Step 1 (Centering): Arrange the data in a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with rows $\mathbf{x}_1, \dots, \mathbf{x}_n$. Define the centering matrix $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ and then the **centered data** is $H\mathbf{X}$.
(its rows are the centered data points $\mathbf{x}_i - \bar{\mathbf{x}}$)

Step 2 (Sample covariance):

$$S = \frac{1}{n}(H\mathbf{X})^\top(H\mathbf{X}).$$

Step 3 (Spectral decomposition):

$$S = U\Lambda U^\top \quad \text{and define } \mathbf{Y} = H\mathbf{X}U \quad (\text{rotate the centered data}).$$

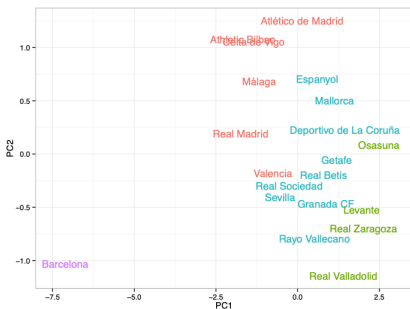
The sample covariance of $\mathbf{Y}$ is $\frac{1}{n}\mathbf{Y}^\top\mathbf{Y} = U^\top S U = \Lambda$ (diagonal! and so $\mathbf{y}_i$ are uncorrelated).

The columns of U are **principal directions** and diagonal entries of Λ are the **explained variances**.

Sorting eigenvalues in decreasing order ranks components by variance captured.

PCA & Unique football styles

“Mille viae ducunt Barcinonem” (A thousand roads lead to Barcelona)



“Searching for a Unique Style in Soccer” (Gyarmati, Kwak & Rodriguez, 2014), studies sequential pass patterns in a team’s passing.

They defined “flow motifs” as statically significant pass subsequences (e.g., $A \rightarrow B \rightarrow C \rightarrow A$).

Applied motif analysis to compare styles across teams. The data are projected to the first two principal components.

Quadratic forms

Quadratic forms

A quadratic form is

$$Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}, \quad A \in \mathbb{R}^{n \times n}.$$

Why symmetric? If $a_{ij} \neq a_{ji}$, the asymmetric part cancels:

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top \frac{1}{2}(A + A^\top) \mathbf{x}.$$

So every quadratic form has a unique symmetric representative.

Examples:

$$Q(x, y) = x^2 + 5xy + 4y^2$$

comes from $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ or equivalently $\begin{pmatrix} 1 & 2.5 \\ 2.5 & 4 \end{pmatrix}$.

Geometry: level sets $\{\mathbf{x} : Q(\mathbf{x}) = c\}$ are ellipses, hyperboloids, linear spaces.

Signs of quadratic forms

For symmetric A :

- **Positive definite (PD)**: $x^\top Ax > 0$ for all $x \neq 0$.
- **Negative definite (ND)**: $x^\top Ax < 0$ for all $x \neq 0$.
- **Semidefinite**: always ≥ 0 (PSD) or ≤ 0 (NSD).
- **Indefinite**: can take both signs.

Why it matters:

- In optimization, a PD Hessian $\Rightarrow$ strict local minimum.
- In economics, concave utility functions $\Rightarrow$ risk aversion.

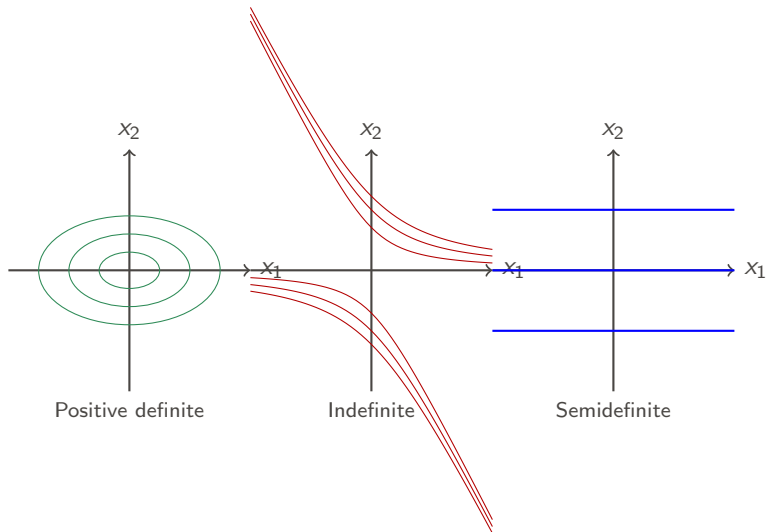
Example:

$$x^\top \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} x = (x - y)^2 \geq 0$$

so the matrix is PSD.

Geometry of quadratic forms

Level sets $\{x : Q(x) = c\}$ illustrate definiteness:



Ellipses $\rightarrow$ PD, Hyperbolas $\rightarrow$ Indefinite, Parallel lines $\rightarrow$ Semidefinite.

Eigenvalue and minor tests

Spectral test (symmetric A):

- PD $\iff$ all eigenvalues > 0 ; ND $\iff$ all < 0 ;
- PSD $\iff$ all ≥ 0 ; NSD $\iff$ all ≤ 0 ;
- Indefinite $\iff$ eigenvalues of different signs.

Sylvester's criterion (leading principal minors D_k):

- PD $\iff D_k > 0$ for $k = 1, \dots, n$;
- ND $\iff (-1)^k D_k > 0$ for $k = 1, \dots, n$.

Example: $A = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & -5 \end{pmatrix}$ has $D_1 = -3$, $D_2 = 5$, $D_3 = -25 \Rightarrow$
negative definite.

Examples of quadratic forms

1. $A = \begin{pmatrix} 1 & 4 & 6 \\ 4 & 2 & 1 \\ 6 & 1 & 6 \end{pmatrix} \Rightarrow$ indefinite (mixed signs of minors).

2. $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix} \Rightarrow$ positive definite.

3. $Q(x, y, z) = xy + yz \Rightarrow$ indefinite (matrix has both positive and negative eigenvalues).

Takeaway: sign of eigenvalues = curvature type.

Chapter 9: Functions of several variables

Many economic and data science models depend on **several variables simultaneously**.

Why this matters: In economics, technologies and preferences are multivariate (production, utility). In data science/ML, high-dimensional loss functions $L(\theta_1, \dots, \theta_d)$ are optimized using gradients/Hessians.

Examples:

- **Economics:** Cobb–Douglas production $Y = K^\alpha L^{1-\alpha}$, utility $U(x, y)$.
- **Data science:** Loss functions $L(\theta)$ with $d \gg 1$ parameters.

Reading: Werner–Sotskov (Ch. 11); Simon–Blume (Chs. 14, 17).

Exercises: 11.11(a), 11.21, 11.22 (Werner–Sotskov).

Basic definitions and examples

What is a multivariable function?

A function of n variables is a map

$$f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{x} = (x_1, \dots, x_n) \mapsto f(\mathbf{x}),$$

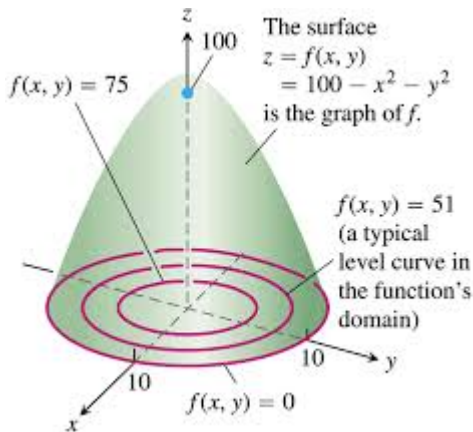
typically with D open. When discussing gradients/tangent planes we assume $f \in C^1$ on D .

- For $n = 2$: the graph $z = f(x, y)$ is a surface in $\mathbb{R}^3$.
- Level curves (contours): $\{(x, y) \in D : f(x, y) = c\}$.
- Sub/superlevel sets: $\{f \leq c\}, \{f \geq c\}$ useful for (quasi-)convexity.
- For $n > 2$: use slices or projections to visualize.

Example: quadratic function

$$f(x, y) = 100 - x^2 - y^2$$

- Graph of f : paraboloid.
- Level curves: concentric circles.



Economic example: Cobb–Douglas

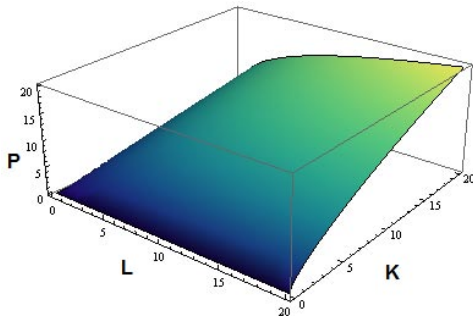
Cobb–Douglas production function

$$P(L, K) = b L^{\alpha} K^{\beta}.$$

P = output, L = labour, K = capital, b = total factor productivity, α, β = output elasticities.

Domain:

$$D = \{(L, K) \in \mathbb{R}^2 : L \geq 0, K \geq 0\}.$$



Multivariate Gaussian density

A random vector $\mathbf{X} \in \mathbb{R}^d$ is **multivariate normal** with mean $\mathbf{m}$ and $\Sigma \succ 0$ if

$$f(\mathbf{x}) = (2\pi)^{-d/2} (\det \Sigma)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{m})\right).$$

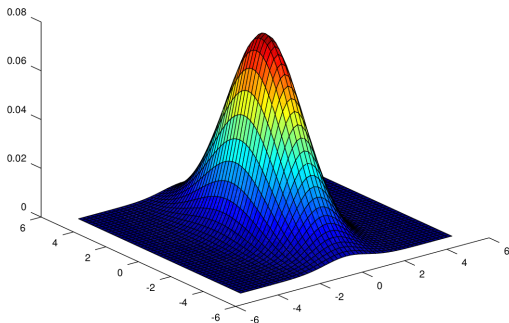
- Parameters: $\mathbf{m} = E(\mathbf{X})$, $\Sigma = \text{Cov}(\mathbf{X}) \succ 0$.
- Depends on *Mahalanobis* distance

$$\|\mathbf{x} - \mathbf{m}\|_\Sigma = \sqrt{(\mathbf{x} - \mathbf{m})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{m})}.$$

- Level sets are ellipsoids $(\mathbf{x} - \mathbf{m})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{m}) = \text{const.}$

Multivariate Gaussian density

Example: $d = 2$, $\mathbf{m} = \mathbf{0}$, variances $\sigma_1^2 = 1$, $\sigma_2^2 = 4$ (zero covariance).



Differentiation

Partial derivatives

Definition

For $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$, the partial derivatives at (x, y) are

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}, \quad f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$

when these limits exist. (We also use more standard notation $\frac{\partial f}{\partial x}(x, y)$, $\frac{\partial f}{\partial y}(x, y)$)

Equivalently, fix y and define $g(x) = f(x, y)$. Then $f_x(x, y) = g'(x)$.

Example (marginal costs): if

$$C(x, y) = 200 + 22x + 16y^{3/2},$$

then $C_x(x, y) = 22$ and $C_y(x, y) = 24\sqrt{y}$.

The gradient and linear approximation

For $\mathbf{x} \in \mathbb{R}^n$, the **gradient** is

$$\nabla f(\mathbf{x}) = (f_{x_1}(\mathbf{x}), \dots, f_{x_n}(\mathbf{x})) \in \mathbb{R}^n.$$

Best linear approximation

If f is a C^1 -function, then for $\mathbf{h} \in \mathbb{R}^n$,

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h} + o(\|\mathbf{h}\|).$$

So the gradient gives the **best linear approximation** of f near $\mathbf{x}$.

Geometry: ∇f is normal to level sets $f(\mathbf{x}) = c$

Why is the gradient the direction of steepest ascent?

Take $\mathbf{h} = t\mathbf{u}$ with $\|\mathbf{u}\| = 1$, $t > 0$ small. Then

$$f(\mathbf{x} + t\mathbf{u}) = f(\mathbf{x}) + t \langle \nabla f(\mathbf{x}), \mathbf{u} \rangle + o(t).$$

The instantaneous rate of change in direction $\mathbf{u}$ is

$$D_{\mathbf{u}}f(\mathbf{x}) := \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{u}) - f(\mathbf{x})}{t} = \langle \nabla f(\mathbf{x}), \mathbf{u} \rangle.$$

By Cauchy-Schwarz,

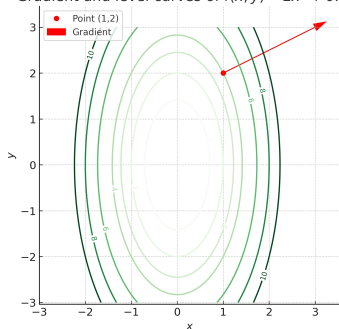
$$|D_{\mathbf{u}}f(\mathbf{x})| \leq \|\nabla f(\mathbf{x})\|,$$

with equality if $\mathbf{u}$ points in the same direction as $\nabla f(\mathbf{x})$.

Conclusion: $\nabla f(\mathbf{x})$ points in the direction of **steepest increase**, $-\nabla f(\mathbf{x})$ in the direction of **steepest decrease**.

Example: gradient and level curves

Gradient and level curves of $f(x, y) = 2x^2 + 0.5y^2$



$$\text{Let } f(x, y) = 2x^2 + \frac{1}{2}y^2.$$

$$\nabla f(x, y) = (4x, y).$$

$$\text{At } (1, 2), \nabla f = (4, 2).$$

Geometry: The gradient is perpendicular to the level curve

$$2x^2 + \frac{1}{2}y^2 = c$$

through $(1, 2)$.

Jacobian and matrix differentiation rules

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The **Jacobian** at $\mathbf{x}$ is

$$JF(\mathbf{x}) = \left[\frac{\partial F_i}{\partial x_j}(\mathbf{x}) \right]_{i=1,\dots,m; j=1,\dots,n} \in \mathbb{R}^{m \times n}.$$

If $m = 1$, $F = f$, then $Jf(\mathbf{x}) = \nabla f(\mathbf{x})^\top$.

First-order approximation

$$F(\mathbf{x} + \mathbf{h}) = F(\mathbf{x}) + JF(\mathbf{x})\mathbf{h} + o(\|\mathbf{h}\|).$$

Quick identities:

1. If $F(\mathbf{x}) = A\mathbf{x}$ with $A \in \mathbb{R}^{m \times n}$: $JF = A$.
2. If $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$: $\nabla f = \mathbf{a}$.
3. If $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$: $\nabla f = (A + A^\top)\mathbf{x}$ ($= 2A\mathbf{x}$ if A symmetric).

Higher partial derivatives

Higher partials are iterated derivatives, e.g.

$$f_{xy}(x, y) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x}(x, y) \right).$$

Young's Theorem:

If f_{xy} and f_{yx} are continuous, then $f_{xy} = f_{yx}$. We write $\frac{\partial^2 f}{\partial x \partial y}$.

Example: $f(x, y) = x^2y + e^{xy}$ gives

$$f_{xy} = 2x + e^{xy}(1 + xy) = f_{yx}.$$

Hessian $\nabla^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n}$

This is a $n \times n$ matrix defined by $(\nabla^2 f(\mathbf{x}))_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$. (symmetric matrix!)

If $f(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$ with A symmetric, then $\nabla^2 f(\mathbf{x}) = 2A$.

Second order Taylor expansion

If $f : D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}^n$ is C^2 then

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h} + \frac{1}{2} \mathbf{h}^\top (\nabla^2 f(\mathbf{x})) \mathbf{h} + o(\|\mathbf{h}\|^2)$$

Note that $\mathbf{h}^\top (\nabla^2 f(\mathbf{x})) \mathbf{h}$ is a quadratic form in $\mathbf{h}$.

If you prefer working with indices (which you should not) this becomes

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\mathbf{x}) h_i + \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) h_i h_j + o(\|\mathbf{h}\|^2).$$

Elements of optimization

Unconstrained optimization

Local optima. $\mathbf{x}_0$ is a local max (min) if $f(\mathbf{x}) \leq (\geq) f(\mathbf{x}_0)$ nearby. If f is differentiable and $\mathbf{x}_0$ is an interior local extremum,

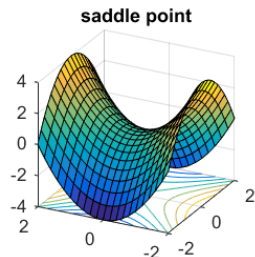
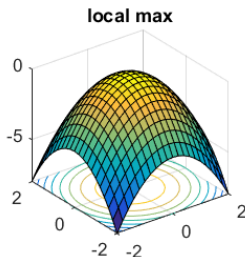
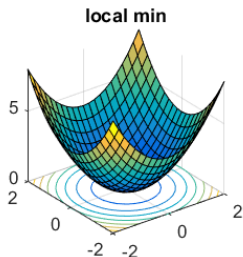
$$\nabla f(\mathbf{x}_0) = \mathbf{0} \quad (\text{first-order condition}).$$

If $f \in C^2$, Taylor expansion at a stationary point $\mathbf{x}_0$ gives

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \frac{1}{2} \mathbf{h}^\top (\nabla^2 f(\mathbf{x}_0)) \mathbf{h} + o(\|\mathbf{h}\|^2),$$

so the sign of the quadratic form $\mathbf{h}^\top (\nabla^2 f(\mathbf{x}_0)) \mathbf{h}$ determines local behavior.

Unconstrained optimization (pictures)



Local optimality: second-order tests

Assume $f \in C^2$ and let $\nabla^2 f(\mathbf{x})$ be the Hessian matrix.

At a stationary point $\mathbf{x}_0$:

- $\nabla^2 f(\mathbf{x}_0)$ positive definite $\Rightarrow$ local minimum.
- $\nabla^2 f(\mathbf{x}_0)$ negative definite $\Rightarrow$ local maximum.
- $\nabla^2 f(\mathbf{x}_0)$ indefinite $\Rightarrow$ saddle.

$n = 2$ test: Let $D_2 = f_{xx}f_{yy} - f_{xy}^2$ at $\mathbf{x}_0$.

$D_2 > 0, f_{xx} > 0 \Rightarrow$ local min,

$D_2 > 0, f_{xx} < 0 \Rightarrow$ local max,

$D_2 < 0 \Rightarrow$ saddle, $D_2 = 0$: inconclusive.

Examples

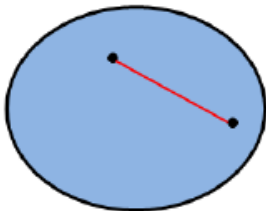
1. $f(x, y) = x^2 - y^2 - xy$. Then $\nabla f = (2x - y, -2y - x)$. The only stationary point is $(0, 0)$. The Hessian $\nabla^2 f(0, 0) = \begin{pmatrix} 2 & -1 \\ -1 & -2 \end{pmatrix}$ is indefinite $\Rightarrow (0, 0)$ is a saddle.
2. $f(x, y) = x^2 + y^4$. Stationary point: $(0, 0)$. $\nabla^2 f(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ is positive semidefinite; $f \geq 0$ so $(0, 0)$ is a global minimum.
3. $f(x, y) = x^3 + y^3$. Stationary point: $(0, 0)$. The Hessian at $(0, 0)$ is 0; the point is a saddle.

Convexity

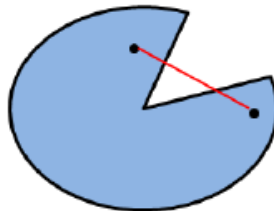
Convex domains

A set $D \subset \mathbb{R}^n$ is **convex** if for any $\mathbf{x}, \mathbf{y} \in D$ and $t \in [0, 1]$, the point $(1 - t)\mathbf{x} + t\mathbf{y} \in D$.

Convex



Non-convex



Convexity, concavity, and global optimality

Definition

On a convex domain $D \subset \mathbb{R}^n$, f is **convex** if

$$f(\theta \mathbf{x} + (1 - \theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta)f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in D, \theta \in [0, 1].$$

Concave: reverse the inequality.

Curvature test (for C^2 functions)

$\nabla^2 f(\mathbf{x}) \succeq 0 \quad \forall \mathbf{x} \in D \iff f$ convex on D , $\nabla^2 f(\mathbf{x}) \preceq 0 \iff f$ concave.
Strict definiteness $\Rightarrow$ strict convexity/concavity.

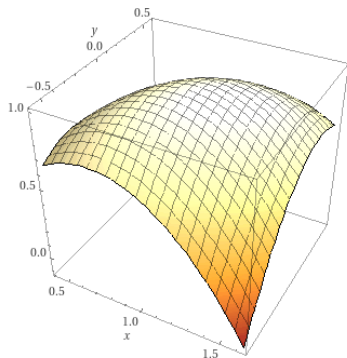
Important fact: If f is strictly convex (concave) then any local minimum (maximum) is a global minimum (maximum).

Example

Let $f(x, y) = 2x - y - x^2 + xy - y^2$. Then

$$\nabla f = (2 - 2x + y, -1 + x - 2y), \quad \nabla^2 f(\mathbf{x}) = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}.$$

$\nabla^2 f(\mathbf{x})$ is **negative definite** $\Rightarrow f$ is strictly concave. The unique stationary point solves $\nabla f = \mathbf{0}$, giving $(x, y) = (0, 1)$, which is a **global maximum**.



plot 2x-y-x^2+xy-y^2 | Computed by Wolfram|Alpha

Economic example: profit maximization

A firm sells products X/Y at 45/55 euros. Revenue $R(x, y) = 45x + 55y$.

Cost

$$C(x, y) = 300 + x^2 + 1.5y^2 - 25x - 35y.$$

Profit $f(x, y) = R(x, y) - C(x, y)$. Then

$$f_x = -2x + 70, \quad f_y = -3y + 90 \Rightarrow (x^*, y^*) = (35, 30).$$

Since $\nabla^2 f(\mathbf{x}) = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$ is negative definite everywhere, f is strictly concave and $(35, 30)$ is the **global maximum**. The maximal profit is $f(35, 30) = 2275$.

Least squares as orthogonal projection

Given data $\mathbf{X} \in \mathbb{R}^{n \times d}$ and response $\mathbf{y} \in \mathbb{R}^n$, the least-squares estimator solves

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^d} \|\mathbf{y} - \mathbf{X}\beta\|^2, \quad \hat{\mathbf{y}} = \mathbf{X}\hat{\beta}.$$

Geometric view (recall Lecture 2): $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$, hence

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0} \quad \Longleftrightarrow \quad (\mathbf{X}^\top \mathbf{X})\hat{\beta} = \mathbf{X}^\top \mathbf{y}.$$

Polynomial regression: A common use of least squares is fitting nonlinear trends by expanding the design matrix $\mathbf{X}$. For instance, with one predictor x , we can set

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{pmatrix},$$

so that the fitted model is $y \approx c_0 + c_1x + \cdots + c_mx^m$.

Ridge regression: stabilizing high-variance fits

When $\mathbf{X}^\top \mathbf{X}$ is ill-conditioned or d is large, add ℓ_2 regularization:

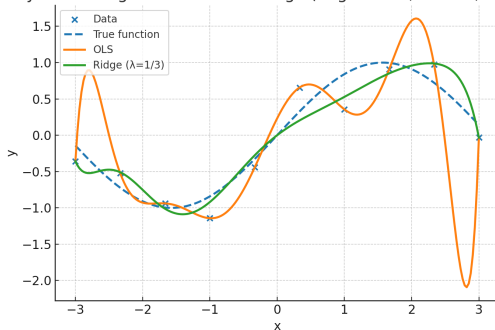
$$\hat{\beta}_\lambda = \arg \min_{\beta} (\|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda \|\beta\|^2) \implies \hat{\beta}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Overfitting vs. ridge: degree-9 polynomial demo

$n = 10$ points from $y = \sin x + \varepsilon$ on $[-3, 3]$, degree 9 polynomial.

OLS (no penalty) vs. Ridge with $\lambda = \frac{1}{3}$.

Polynomial Regression: OLS vs Ridge (degree = 9, $n = 10$, $\sigma = 0.2$)



OLS wiggles violently between sparse points; ridge damps the high-frequency coefficients and tracks the signal more smoothly.

Modern applied examples (multivariable)

- Portfolio risk (mean–variance):

$$f(\mathbf{w}) = \mathbf{w}^\top \Sigma \mathbf{w}, \quad g(\mathbf{w}) = \mu^\top \mathbf{w}, \quad \mathbf{w} \in \mathbb{R}^n, \quad \sum_i w_i = 1, \quad w_i \geq 0.$$

- Logistic regression (binary choice):

$$J(\beta) = \frac{1}{n} \sum_{i=1}^n \left(\log(1 + e^{\mathbf{x}_i^\top \beta}) - y_i \mathbf{x}_i^\top \beta \right) \quad (\text{convex in } \beta).$$

- CES utility/production:

$$U(x) = \left(\sum_{i=1}^n \alpha_i x_i^\rho \right)^{1/\rho}, \quad P(L, K) = A(\theta) L^\alpha K^\beta.$$

Numerical optimization

Gradient descent

Goal: minimize $f(\mathbf{x})$.

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \nabla f(\mathbf{x}_t).$$

Pieces you pick:

- **Step size** η_t : constant, diminishing, or via backtracking.
- **Stop** when $\|\nabla f(\mathbf{x}_t)\|$ small.

In practice: a good η_t schedule matter a lot.

GD on least squares (closed form vs iterations)

Least squares problem has a closed form solution. This still requires inverting a potentially large matrix $\mathbf{X}^\top \mathbf{X}$. GD gives an alternative way to find a solution.

$$f(\beta) = \frac{1}{n} \|\mathbf{X}\beta - \mathbf{y}\|^2, \quad \nabla f(\beta) = \frac{2}{n} \mathbf{X}^\top (\mathbf{X}\beta - \mathbf{y}).$$

GD update:

$$\beta_{t+1} = \beta_t - \eta \frac{2}{n} \mathbf{X}^\top (\mathbf{X}\beta_t - \mathbf{y}).$$

Ridge:

$$f_\lambda(\beta) = \frac{1}{n} \|\mathbf{X}\beta - \mathbf{y}\|^2 + \lambda \|\beta\|_2^2, \quad \nabla f_\lambda = \frac{2}{n} \mathbf{X}^\top (\mathbf{X}\beta - \mathbf{y}) + 2\lambda\beta.$$

Closed form exists $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, but GD scales better to huge n, p or streaming data.

When to use second order methods?

In general, we update $\mathbf{x}_t$ as

$$\mathbf{x}_{t+1} := \arg \min f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{x} \rangle + \frac{1}{2}(\mathbf{x} - \mathbf{x}_t)^\top \mathbf{K}(\mathbf{x} - \mathbf{x}_t).$$

If $\mathbf{K} = I_n$, we recover gradient descent.

If $\mathbf{K} = \nabla \nabla^\top f(\mathbf{x}_t)$, we get the **Newton method**.

- **Newton:** $\mathbf{x}_{t+1} = \mathbf{x}_t - H^{-1}(\mathbf{x}_t) \nabla J(\mathbf{x}_t)$ (fast near solution, expensive to form/solve).
- **Quasi-Newton (e.g., BFGS, LBFGS):** approximate H^{-1} from gradients only; great for medium scale convex problems.
- **Takeaway:** for huge data/models use (S)GD; for smaller smooth convex problems, quasi-Newton shines.

Constrained optimization

Constrained optimization: Lagrange and KKT (teaser)

We want to maximize/minimize $f(\mathbf{x})$ subject to constraints:

$$g_i(\mathbf{x}) = 0 \quad (\text{equalities}), \quad h_j(\mathbf{x}) \leq 0 \quad (\text{inequalities}).$$

Lagrangian:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_i \lambda_i g_i(\mathbf{x}) + \sum_j \mu_j h_j(\mathbf{x}).$$

KKT conditions (when applicable):

1. **Stationarity:** $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = 0$.
2. **Primal feasibility:** $g_i(\mathbf{x}^*) = 0, \quad h_j(\mathbf{x}^*) \leq 0$.
3. **Dual feasibility:** $\mu_j^* \geq 0$.
4. **Complementary slackness:** $\mu_j^* h_j(\mathbf{x}^*) = 0$.

Constrained optimization in 3D (example)

Problem: maximize

$$f(x, y, z) = xyz \quad \text{subject to} \quad g(x, y, z) = x + y + z - 1 = 0, \quad x, y, z \geq 0.$$

Lagrangian:

$$\mathcal{L}(x, y, z, \lambda, \mu_1, \mu_2, \mu_3) = xyz + \lambda(1 - x - y - z) - \mu_1 x - \mu_2 y - \mu_3 z.$$

Stationarity:

$$\partial_x \mathcal{L} = yz - \lambda - \mu_1 = 0,$$

$$\partial_y \mathcal{L} = xz - \lambda - \mu_2 = 0,$$

$$\partial_z \mathcal{L} = xy - \lambda - \mu_3 = 0.$$

Complementary slackness: $\mu_1 x = \mu_2 y = \mu_3 z = 0$.

Solution and geometry

At an interior solution ($x, y, z > 0$), complementary slackness
 $\Rightarrow \mu_1 = \mu_2 = \mu_3 = 0$.

Simplified system:

$$yz = \lambda,$$

$$xz = \lambda,$$

$$xy = \lambda,$$

$$x + y + z = 1.$$

Deduction: $yz = xz = xy \Rightarrow x = y = z$. Constraint $\Rightarrow x = y = z = \frac{1}{3}$.

Maximizer: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, value $f^* = \frac{1}{27}$.

Geometric intuition:

- Constraint $x + y + z = 1$ is a plane.
- Level sets $xyz = c$ are curved surfaces.
- At optimum: $\nabla f = (yz, xz, xy)$ is parallel to $\nabla g = (1, 1, 1)$ the plane just *touches* the surface.